

IAS PRELIMS/MAINS



SCIENCE & TECHNOLOGY



A COMPREHENSIVE COVERAGE

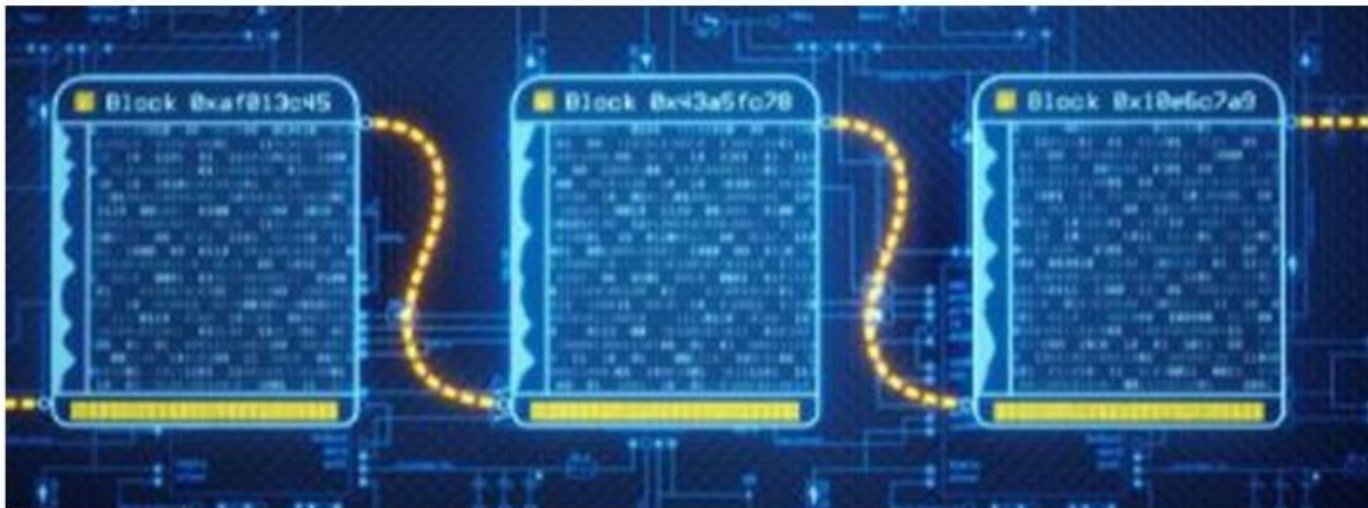
MODULE - 29

INFORMATION & COMMUNICATION TECHNOLOGY (ICT) - 10

BLOCK CHAIN TECHNOLOGY

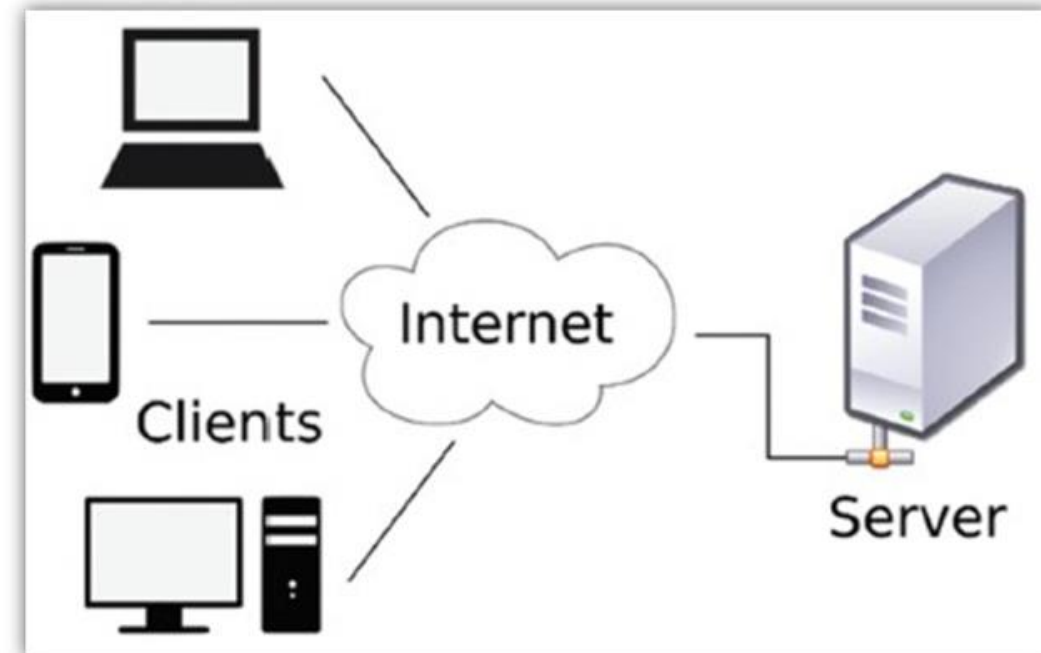
INTRODUCTION

- Blockchain is the technology for all future financial transactions all over the globe and much more.
- Blockchain is the record keeping technology behind Bitcoin and the other cryptocurrencies.
- A block contains digital information stored in a chain which is in public database.
- Blockchain is a distributed, decentralized, public ledger, which stores information about transactions and the participators



DECENTRALIZED PROCESSING OF TRANSACTIONS

- In a centralized system all the data is stored in one location only which can be accessed only by directly interacting with the entity, example a bank. If you want any information about your account, the only way to have it is by interacting with the bank.
- This is the traditional client-server architecture.
- Here, a request is sent to the server and the server responds with the requested information.



DECENTRALIZED PROCESSING OF TRANSACTIONS

- This architecture poses serious risks. Since all the information is stored in the server it becomes the target of hackers.
- Also, upgradation of the server software poses a problem to the clients, as they cannot use it while it is being upgraded.
- Also, if malware gains access to the server, it can corrupt the entire database. So, Decentralized Systems are better.
- In decentralized systems, the data is not stored in one location, but is distributed throughout the network. So, it is not necessary to interact with any one particular server for getting information, the information can be obtained from any computer in the network.

Centralized vs Decentralized vs Distributed Network: An Overview



Centralized Network
All the nodes are connected under a single authority



Decentralized Network
No single authority server controls the nodes, they all have individual entity



Distributed Network
Every node is independent and interconnected with each other

HOW DOES THE BLOCKCHAIN TECHNOLOGY WORK?

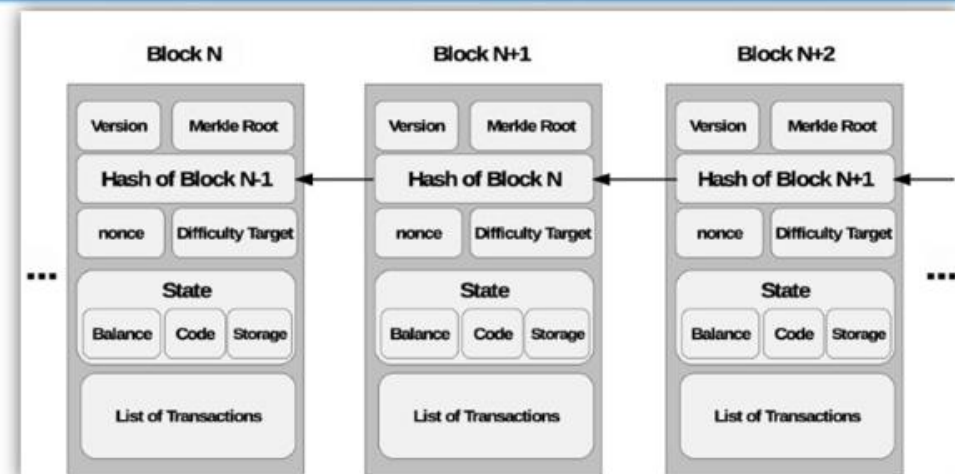
- The Blockchain, being a decentralized and distributed architecture has no centralized points of vulnerability that hackers can exploit.
- So, Blockchain architecture is much more secure than the centralized system which uses the traditional 'username-password' system.
- Blockchain is the concept underlying cryptocurrencies which have no physical form and only exist in the network.
- Whenever a new transaction occurs, a new block is created with new data which is added to the chain. The new block becomes visible to all the computers connected to the network, making it completely transparent. Only the username or digital signature of the user is visible.
- So, Blockchain technology is a publicly distributed ledger with no single entity controlling it. This series of recorded transactions stores information in a chain-like system.



BLOCK CHAIN TECHNOLOGY

HOW DOES THE BLOCKCHAIN TECHNOLOGY WORK?

- Whenever a new block is created, it is given a unique, identifying code called a 'hash' by which it is distinguished from all the other blocks. Each block in the chain is also given an exact timestamp when it is added to the chain. The timestamp proves that the transaction data existed when the block was published in order to get into its hash
- The hash is an algorithm and it lies at the heart of the Blockchain technology. Hashing converts an input string of any length to an output of fixed length.
- For cryptocurrencies, transactions are taken as input which the hashing algorithm converts into an output of fixed length (alphanumeric code).
- In a Blockchain, new blocks are continuously added to the end of the chain, the position of the block is called its 'Height'.
- Once a block is added to the chain, it becomes impossible to alter its contents because each block contains a hash of the previous block, a timestamp and the transaction data.



BLOCK CHAIN TECHNOLOGY

HOW DOES THE BLOCKCHAIN TECHNOLOGY WORK?

- Any editing of the block's information would also change the block's hash. But the next block in the chain will still contain the old hash and a hacker wanting to erase that must update every single block after it on the Blockchain, which is impossible. This is what makes Blockchain so secure.
- Also, the Blockchain is controlled by a whole community of users, and every addition to the chain is carefully tracked. This means that no organization, person can make changes to the information recorded once it's there.
- Another way the blockchain is secured is by requiring new computers aspiring to join the network to prove themselves by passing tests.
- Bitcoin uses a method in which a computer aspiring to join the blockchain must solve a complex math problem which is very tough. To solve the problem a person must spend a significant amount of money (for large computational power and electricity). This cost would prohibit hackers from joining the network to steal financial data.



OTHER APPLICATIONS OF BLOCKCHAIN TECHNOLOGY

- By allowing digital information to be distributed but not altered, Blockchain technology has created the backbone for a new internet. Blockchain ensure accuracy of data and improves privacy and security.
- Not only cryptocurrencies, the Blockchain technology can be used in many other fields. Banks and financial institutions will be the largest users of this technology to securely store all transactional data.
- Healthcare industry can also immensely benefit from this system to securely store patient's data and medical records.
- Most importantly Blockchain technology can be used in elections. Voting with Blockchain can eliminate election frauds by making each vote store as a block on the Blockchain making it impossible to tamper with.



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